Revision 2021 Engineering Handbook

Material Science and Metrology

Course code 3022. Program Core for Diploma in Mechanical Engineering / Tool and Die Engineering / Manufacturing Technology.

Semester 3/3/3 | Credits 3 | Periods per week 3 (L:2,T:1,P:0) | Periods per semester 45.

> Ithu simple note alla. Materials selection, testing, heat treatment, measurement accuracy, gauges and machine tool testing full handbook style-il arrange cheythu.

Course Overview

Objectives

1. Identify significance of engineering materials, properties and applications.
2. Understand engineering measurements, measuring techniques, instruments and machine tool testing.

Prerequisites

Topic	Course	Semester
Basic ideas of applied physics	Applied Physics I and II	1 and 2
Classification of engineering materials	Applied Chemistry	1

Course Outcomes

CO	Outcome	Hours	Level
CO1	Explain crystal structure, engineering materials, steels and ferrous alloys.	11	Understanding
CO2	Explain failure, testing and heat treatment.	11	Understanding
CO3	Explain measuring instrument characteristics and force/torque measurement.	11	Applying
CO4	Explain measuring instruments and machine tool inspection/testing.	12	Applying

Redesigned Syllabus Roadmap

CO1 Materials

BCC, FCC, HCP, iron-carbon diagram, TTT, steels, cast iron and alloy steels. Draw crystal cells and TTT curves.

CO2 Testing

Failure, destructive testing, NDT, heat treatment and non-ferrous alloys. Draw stress-strain curve and NDT flow.

CO3 Metrology

Measurement standards, errors, calibration, accuracy, precision, force, torque, load cell and strain gauge.

CO4 Instruments

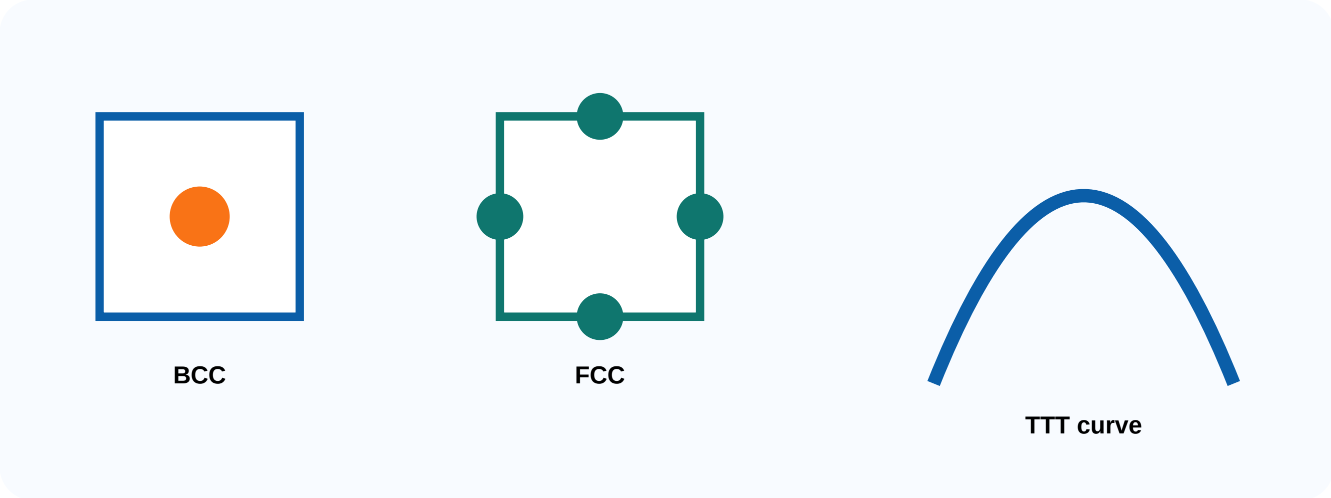
Gauges, comparators, surface texture, CMM, sine bar, autocollimator and machine tool testing.

Module 1 - Crystal Structures and Ferrous Materials

Unit cell is the smallest repeating block of a crystal. Space lattice is the regular 3D repetition of crystal points. BCC has 2 effective atoms per cell, FCC has 4, and HCP is close-packed.

Iron-carbon diagram explains ferrite, austenite, cementite, pearlite, eutectoid steel and cast iron range. TTT diagram shows transformation of austenite with time and temperature.

Pig iron, cast iron, wrought iron, mild steel, HSS, silicon steel, spring steel and stainless steel are selected by composition, hardness, corrosion resistance, machinability and use.



Expected questions

1. Draw BCC, FCC and HCP.
2. Explain importance of iron-carbon diagram.
3. Explain TTT diagram.
4. Classify steels based on carbon content.
5. Explain cast iron, wrought iron and alloy steels.

Module 2 - Failure, Testing and Heat Treatment

Ductile fracture gives plastic deformation before failure; brittle fracture is sudden. Notch sensitivity, fatigue and creep must be understood for shafts, gears, brackets and heated components.

Destructive testing includes tensile, compression, Brinell, Rockwell and Vickers hardness tests. NDT includes visual inspection, magnetic particle inspection, liquid penetrant testing, ultrasonic inspection and radiography.

Heat treatment processes include annealing, normalizing, hardening, tempering, martempering, austempering and case hardening such as carburizing, nitriding and cyaniding. Non-ferrous alloys include brass, bronze, duralumin, hindalium, magnelium and magnesium alloys.

> Solved example: use liquid penetrant test for surface crack in non-magnetic machined component. Use annealing when improved ductility and machinability are required.

Module 3 - Metrology and Force/Torque Measurement

Measurement is comparison with a standard. Direct measurement reads value directly; indirect measurement calculates value from related quantities. Generalized measuring system: input, sensor, signal conditioning, display/record and calibration feedback.

Accuracy is closeness to true value; precision is closeness among repeated readings. Sensitivity, repeatability, range, threshold, hysteresis and calibration are core terms. Errors may be systematic or random.

Force measurement uses spring balance, proving ring, strain gauge and load cell. Torque is turning moment: T = F x r.

> Solved example: 50 N force at 0.25 m radius gives torque = 50 x 0.25 = 12.5 N m.

Module 4 - Gauges, Comparators and Machine Tool Testing

Gauges check limit condition. Plug gauge checks holes, ring gauge checks shafts, snap gauge checks external dimensions, feeler gauge checks gaps and thread pitch gauge checks threads. In GO-NOGO gauge, GO should pass and NOGO should not pass.

Comparators compare component dimension with a standard and magnify small deviations. Surface texture includes roughness, waviness and lay. CMM measures 3D coordinate points using a probe.

Angular instruments include universal bevel protractor, clinometer, sine bar, spirit level, autocollimator and angle gauges. Machine tool testing includes straightness, parallelism, squareness, coaxiality, roundness, runout and alignment.

> Solved example: for sine bar length 100 mm and angle 30 degree, h = L sin theta = 100 x 0.5 = 50 mm.

Formula / Rule Bank

Stress

sigma = P / A

Strain

epsilon = delta L / L

Torque

T = F x r

Sine bar

sin theta = h / L

Runout

max dial reading - min dial reading

GO-NOGO

GO passes, NOGO does not pass

Practice and Quick Revision

1. FCC has face-centre atoms.
2. Radiography is NDT.
3. Accuracy means closeness to true value.
4. Torque is F x r.
5. Sine bar uses sin theta = h/L.
6. Comparator compares with a standard.

Industrial relevance: material selection for shafts, gears, brackets, tools and machine beds; metrology for inspection room, tool room, quality control, calibration and machine maintenance.

Fresh Model Question Paper

Time: 3 Hours. Maximum Marks: 100.

1. Define unit cell and space lattice.
2. List BCC, FCC and HCP.
3. Explain TTT diagram.
4. Differentiate ductile and brittle fracture.
5. List NDT methods.
6. Define accuracy and precision.
7. Explain generalized measuring system.
8. Explain hardness tests.
9. Solve sine bar problem for L=100 mm and angle=30 degree.
10. Explain gauges, comparators and machine tool testing.

Complete Answers and References

1. Unit cell is smallest repeating crystal block; space lattice is repeated 3D arrangement.
2. BCC has 2 atoms, FCC 4 atoms and HCP close-packed hexagonal arrangement.

3. TTT diagram helps choose cooling rate and predict transformation products.
4. Ductile fracture has plastic deformation; brittle fracture is sudden.
5. NDT methods: visual, MPI, LPT, ultrasonic and radiography.
6. Accuracy is closeness to true value; precision is closeness of repeated readings.
7. Generalized measurement system contains input, sensor, conditioning, display and calibration.
8. Brinell uses ball, Rockwell uses depth, Vickers uses diamond pyramid.
9. h = 100 x sin 30 = 50 mm.
10. Machine testing checks straightness, parallelism, squareness, roundness, runout and alignment.

References

Collett and Hope; Chapman; Callister; R. K. Rajput; Beckwith, Marangoni and Lienhard; K. J. Hume.
Online: BIPM, NIST, NPTEL and free video lectures.